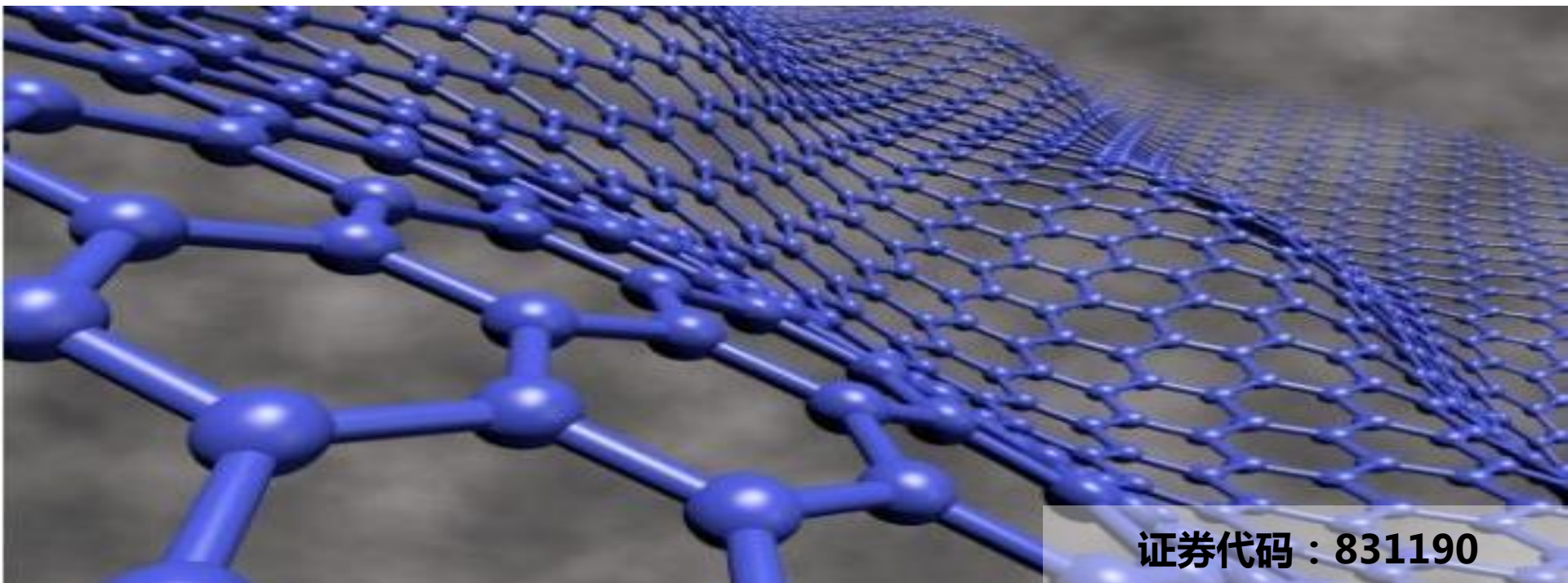


Large scale of rGO manufacture and its application in anticorrosion coating



证券代码 : 831190

Graphene & 2D Materials USA

常州第六元素材料科技股份有限公司
The Sixth Element (Changzhou) Materials Technology Co.,Ltd.

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The Sixth Element Inc.

常州第六元素材料科技股份有限公司
The Sixth Element (Changzhou) Materials Technology Co.,Ltd.

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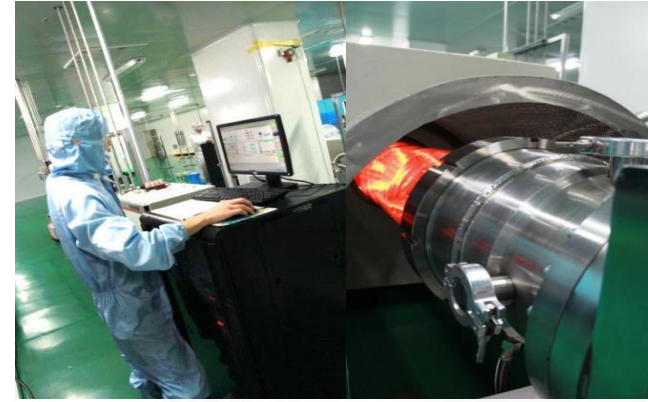
We are a leading graphene manufacturer



50, 300 t/year production lines
for graphene oxide
and graphene powder



**115 employees and 25 in
R&D**



**Large-scaled (15K pieces/day)
CVD-grown graphene
on various substrates**





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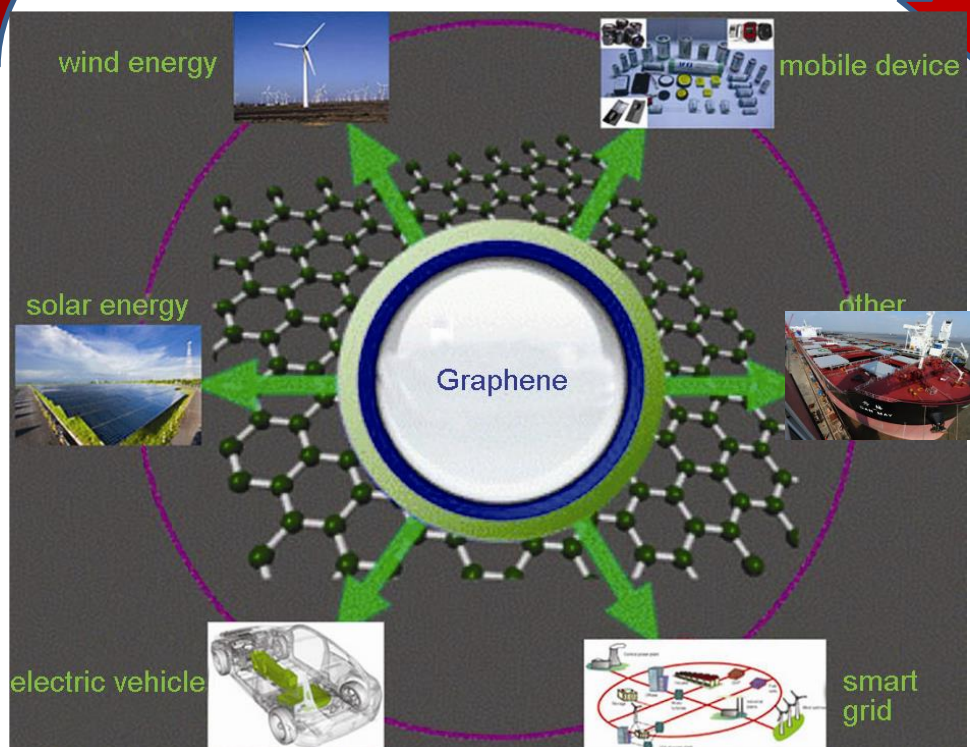
Technology Innovation

Solving the key scientific and technical problems in graphene commercialization

Low Cost Mass Production

Great properties:

- ✓ High Strength
- ✓ High Thermal Conductivity
- ✓ High Electro Mobility
- ✓ High BET
- ✓ Flexibility of Chemical Alternation



Application requirements:

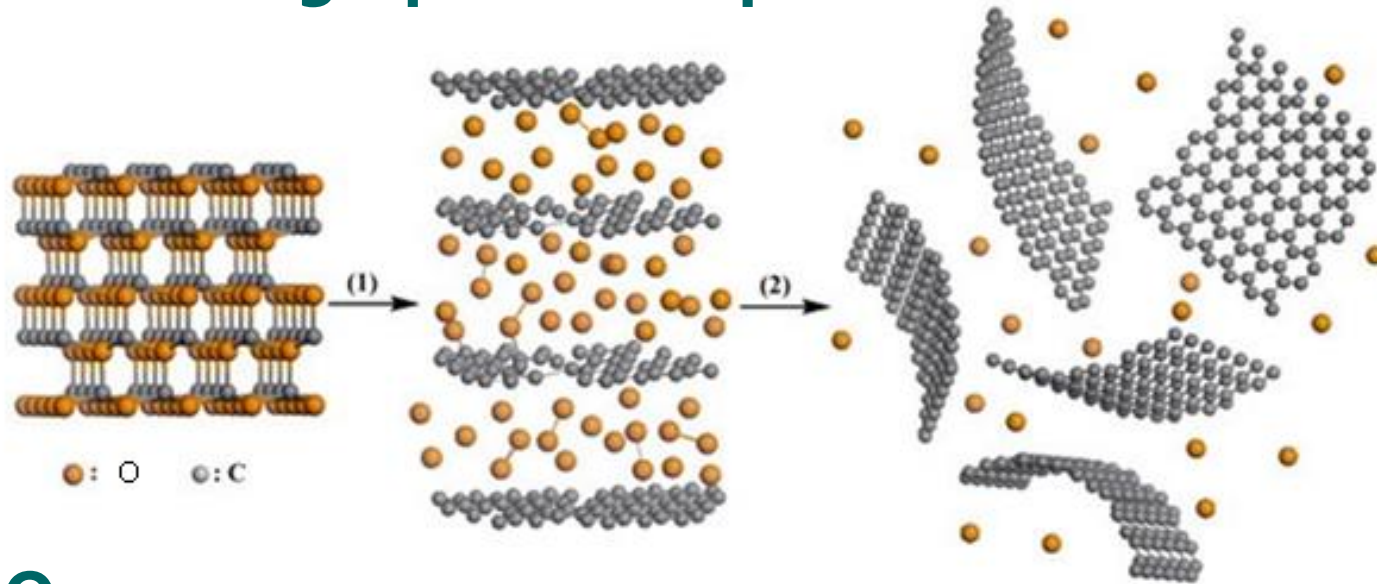
- High Quality
- Low Cost
- Environmental Friendly
- Precise control
- Performance Improvement
- Compatible with Current Technologies



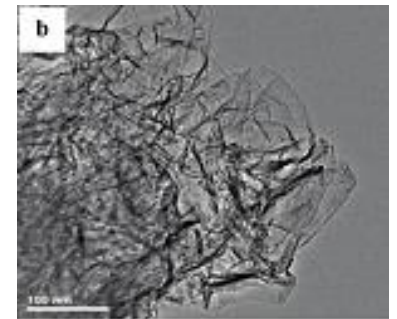
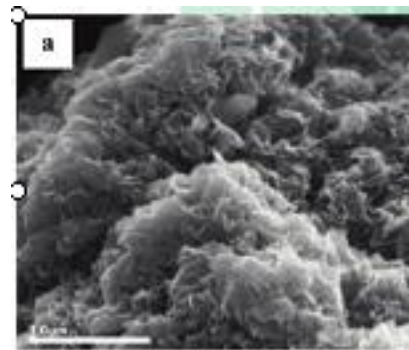
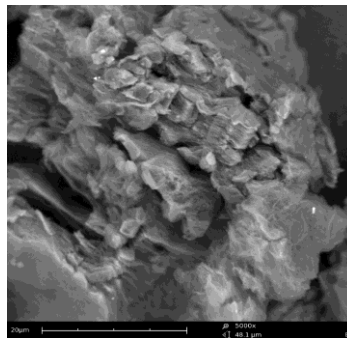
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Technical Path

Expandable graphite + improved Hummers method



rGO





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Product Series



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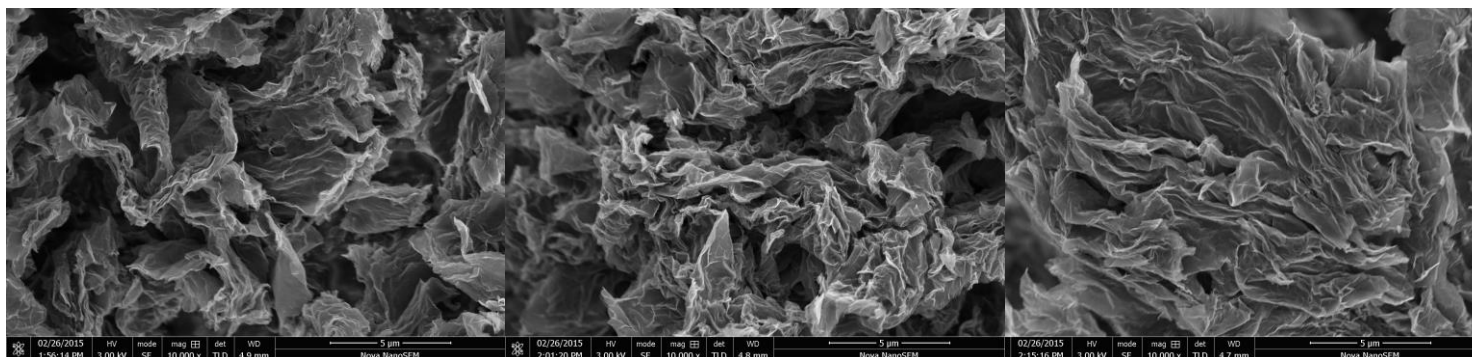
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Characteristics of graphene-flakes

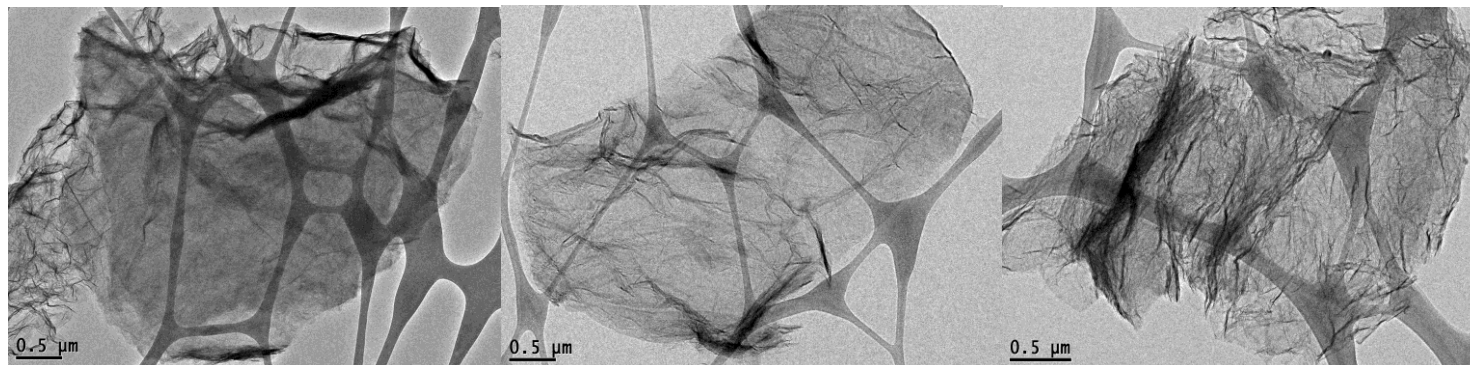
MICROSCOPIC MORPHOLOGY

■ SEM、TEM

SEM Image Comparison of lab test (left) 、 pilot line (middle) and production line (right)



TEM Image Comparison of lab test (left) 、 pilot line (middle) and production line (right)





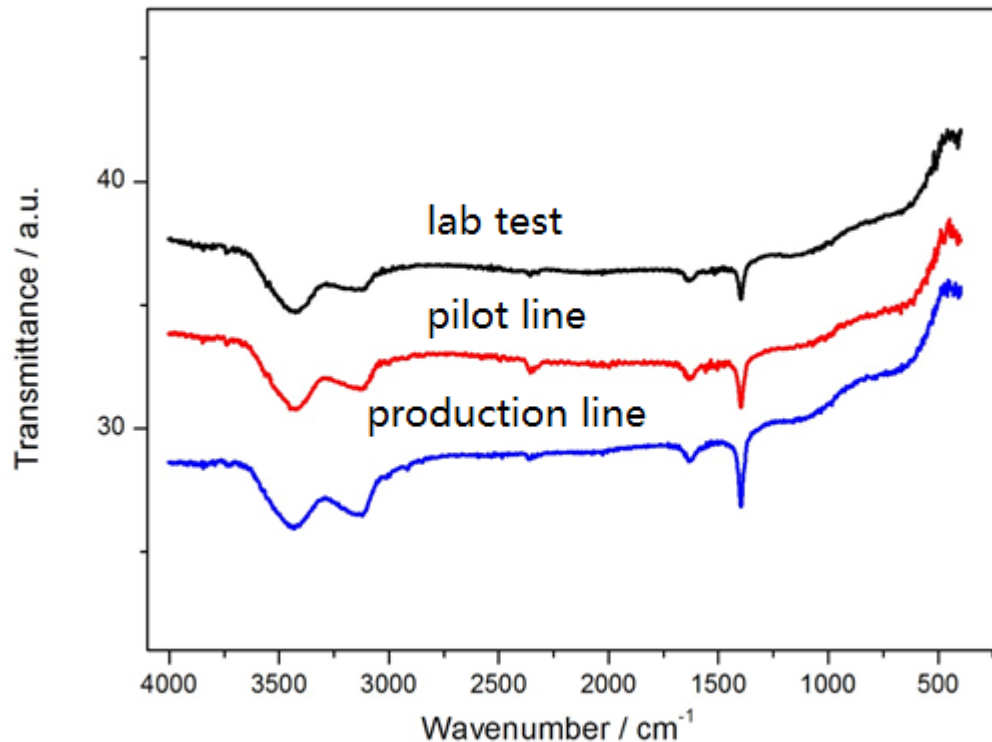
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Characteristics of graphene-flakes

CHEMICAL CHARACTERISTICS

■ IR CHARACTERIZATION

Comparison of infrared spectra



From IR characterization we can see the functional groups in the lab 、 pilot line and production line samples are consistent.



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Characteristics of graphene-flakes

BATCH CONSISTENCY

■ BET TESTS

Consecutive 6 batches BET test results

批次	1	2	3	4	5	6
比表面积 (m^2/g)	200.7	203.4	204.8	207.5	206.6	210.1
测试条件	Nova 2000e BET tester , 260°C degas 2h , absorption pressure 0.05~0.30 , test points.					

■ POWDER CONDUCTIVITY TESTS

Consecutive 6 batches powder conductivity test results

批次	1	2	3	4	5	6
粉末电导率 (S/m)	5539	5308	5551	5265	5705	5479
测试条件	ST2722 type powder conductivity tester , $D_{50} \leq 10\mu\text{m}$, water content $\leq 1.50\%$, test pressure 14MPa.					



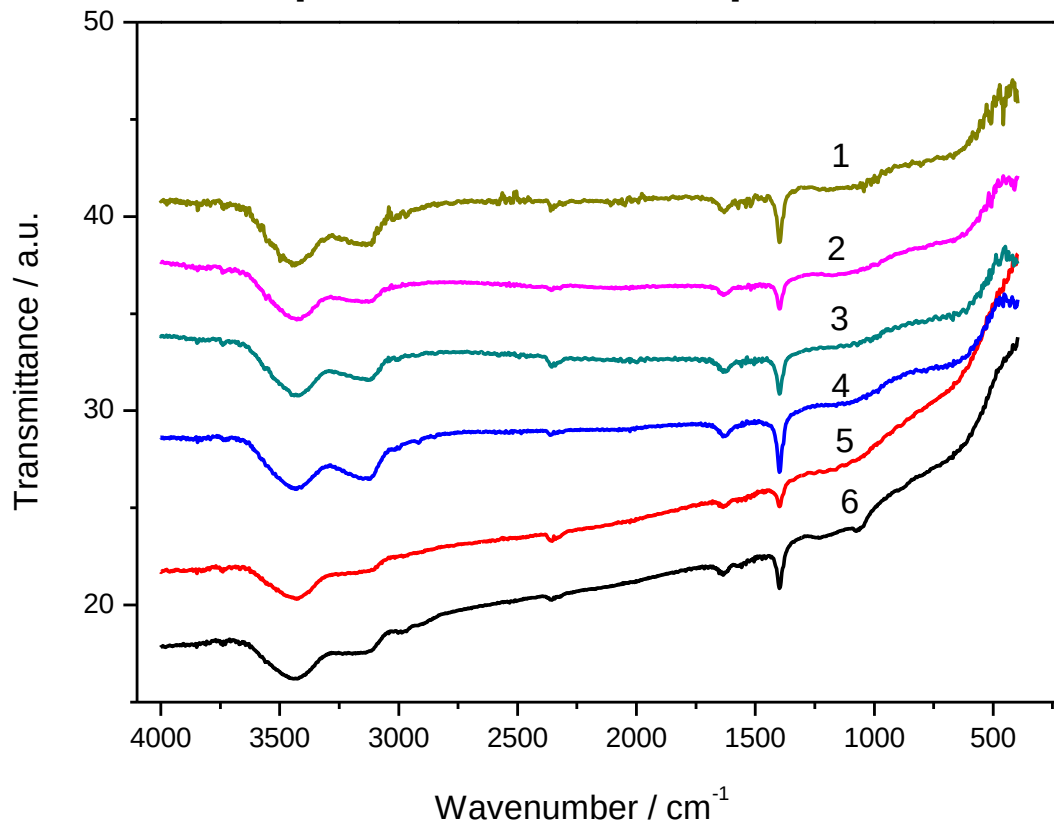
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Characteristics of graphene-flakes

BATCH CONSISTENCY

■ IR CHARACTERIZATION

Comparison of infrared spectra



6 batches of graphene-flakes all contains of a small amount of hydroxyl and carboxyl groups , batch to batch are consistent.



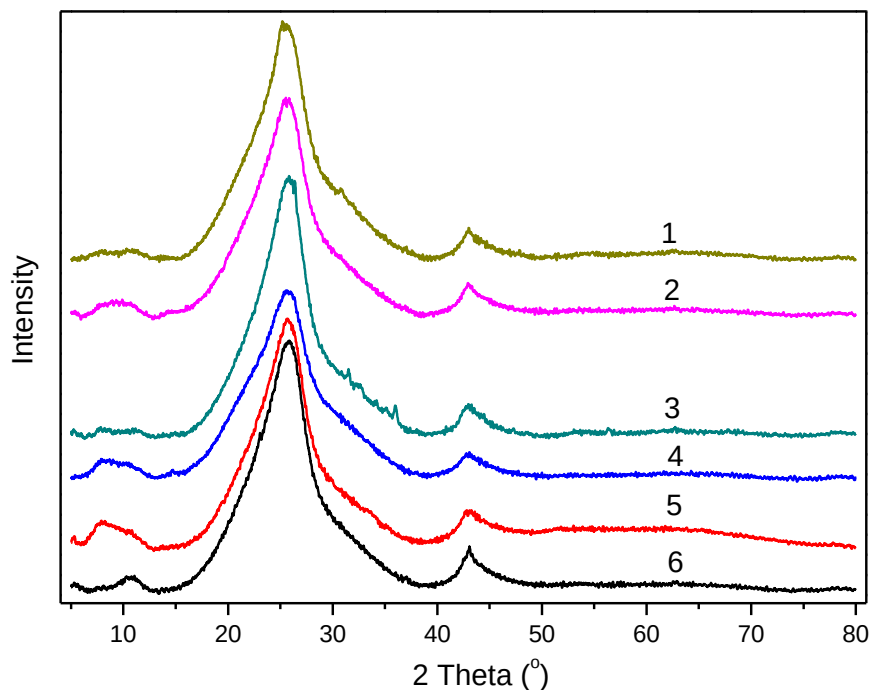
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Characteristics of graphene-flakes

BATCH CONSISTENCY

■ XRD CHARACTERIZATION

Comparison of XRD



6 batches of graphene-flakes all show wide typical peaks at $20^\circ \sim 30^\circ$ in XRD spectrum, the positions and the widths are consistent.



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Hempel--AvantGuard

In 2007, Hempel lab in Danmark discovers: only one third of the zinc in the zinc rich epoxy primer is taking an effect of anode protection during electrochemical corrosion. To resolving this problem their R&D team invest large amount of effort and AvantGuard is the result of this effort.

Salt Spray Test



Zinc epoxy
without
AvantGuard®
technology



Zinc epoxy
with
AvantGuard®
technology
(986 hours)



Full system
without
AvantGuard®
technology
(1440 hours)



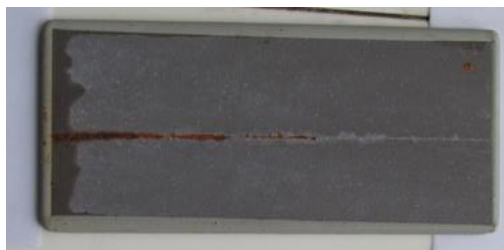
Full system
with
AvantGuard®
technology
(1440 hours)



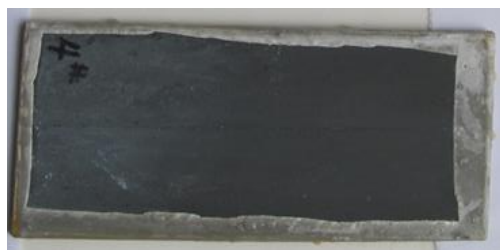
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for anti-corrosive epoxy primer

% Zinc powder	% Graphene	Salt spray test	Impact resistance	Adhesion
20	0	48 h	-	-
70	0	624 h	50 cm	6 MPa
20	1	2500 h	60 cm	15 MPa

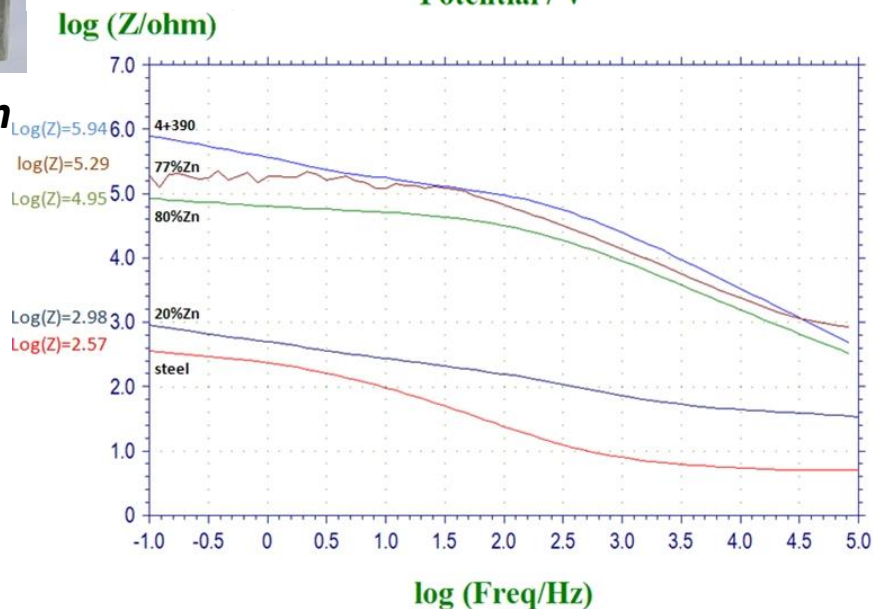
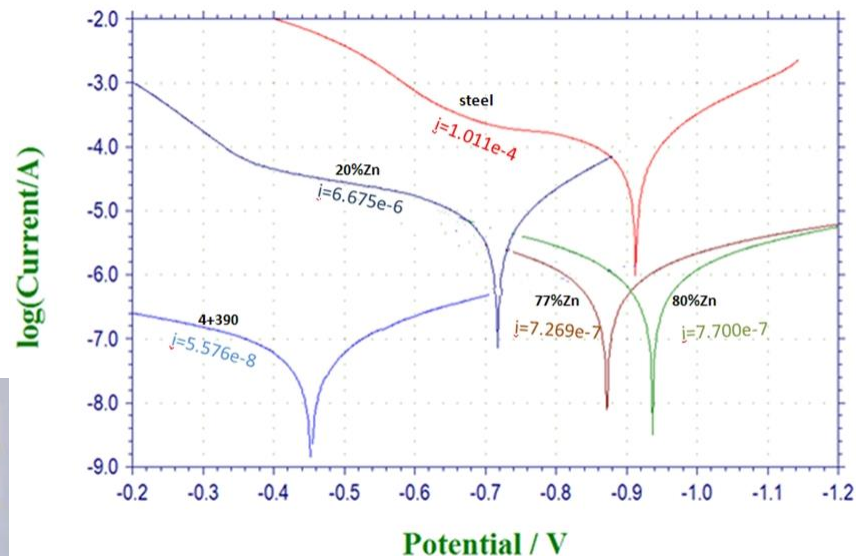


70 % Zn @ 810h



20% Zn + 1% G @ 2509h

applied to offshore wind turbine

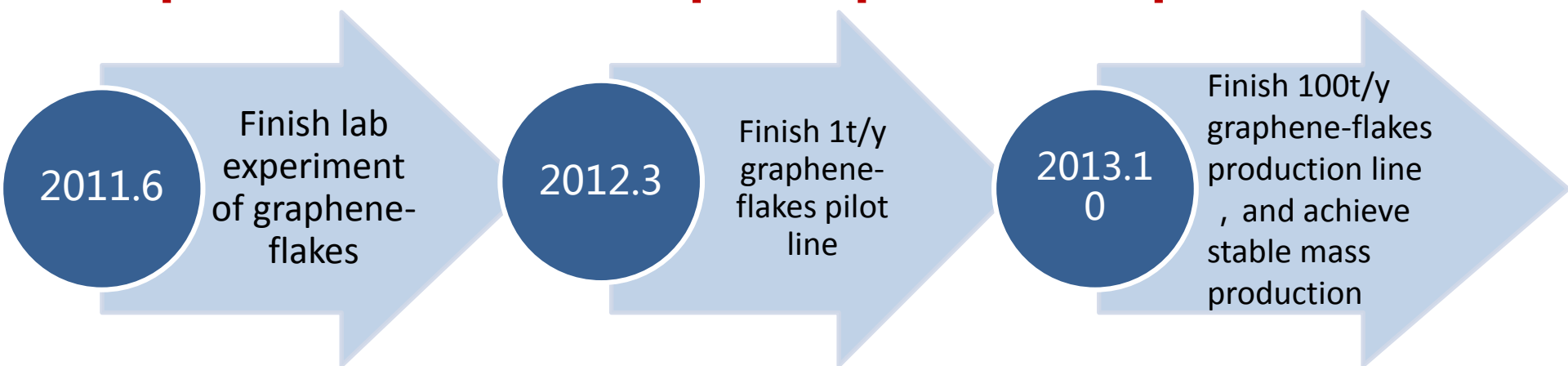




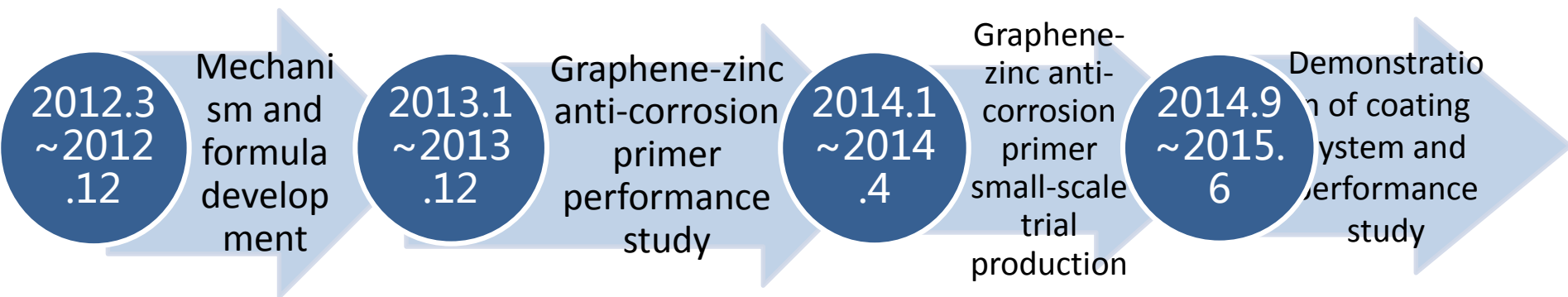
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Project development process

Graphene-flakes develop and productize process



Graphene-zinc anti-corrosion primer and its anticorrosive coating system develop process

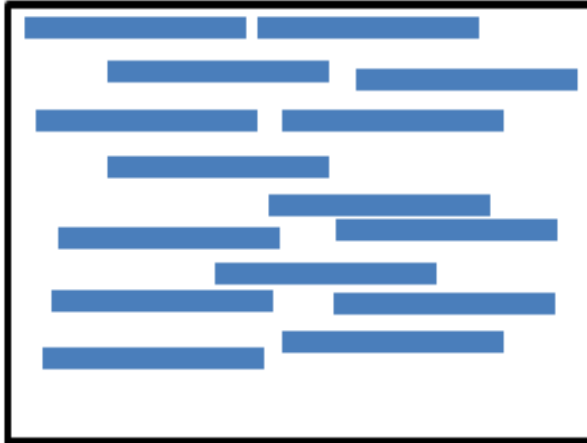




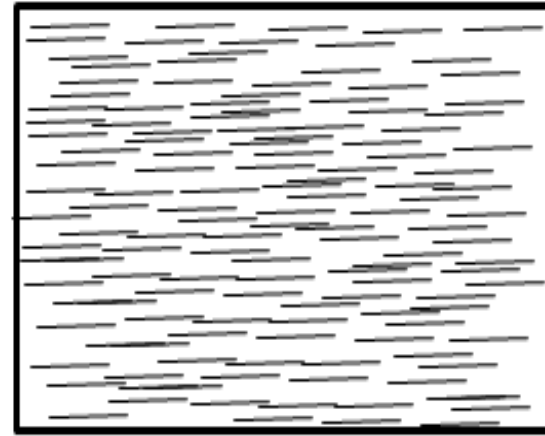
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Graphene-zinc primer anti-corrosion mechanism

Shielding and Connecting



contain 30% glass flakes



contain 1.5% graphene-flakes

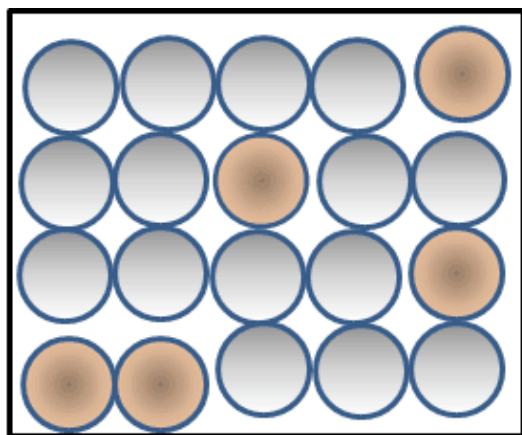
- Using epoxy glass flake coating as a reference, dry film contains 325 mesh size glass scale for 30% , the coating has very good shielding effect and the quantities of glass flakes in the order of 10^9 .
- In the graphene-zinc primer, when the dry film contain 1.5% graphene flaks, the quantities of particles in the order of 10^{12} which is 1000 times of the glass flakes。 Therefore, the graphene-zinc primer have much better shielding and concocting effect.



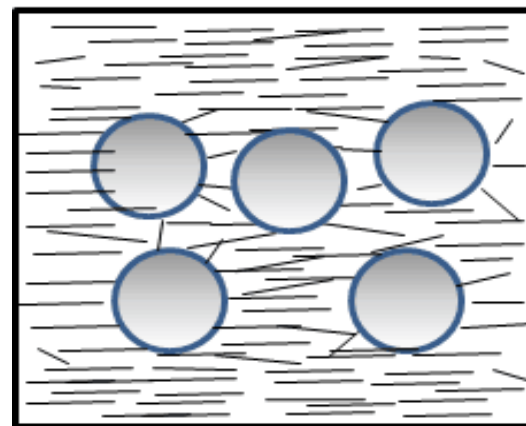
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Graphene-zinc primer anti-corrosion mechanism

Shielding and Connecting



contain 80% Zinc



contain 25% zinc+1% graphene

- In the dry film of the graphene-zinc primer, the graphene flakes connecting each other and forms conductive network.
- The graphene conductive network, connecting the zinc and forms the circuitry to active the zinc for anode protection which increases its usage.



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Cost Analysis

According to the coating specification requirements for offshore wind power tower, primer coating thickness is 60um, the cost analysis as follow : rGO priced at 1000RMB/kg (current commercial price) , zinc powder priced at 18RMB/kg , zinc rich epoxy primer coating costs 3.63RMB/m² , while graphene-zinc anti-corrosion primer coating costs 3.19RMB/m². the detailed calculation as follow:

Coating Name	Unit Cost RMB/L	Density g/mL	Unit coating Area m ² /L	Unit Coating Cost RMB/m ²
zinc rich epoxy primer	40.80	2.5	11.24	3.63
graphene-zinc anti-corrosion primer	40.89	2.1	12.82	3.19



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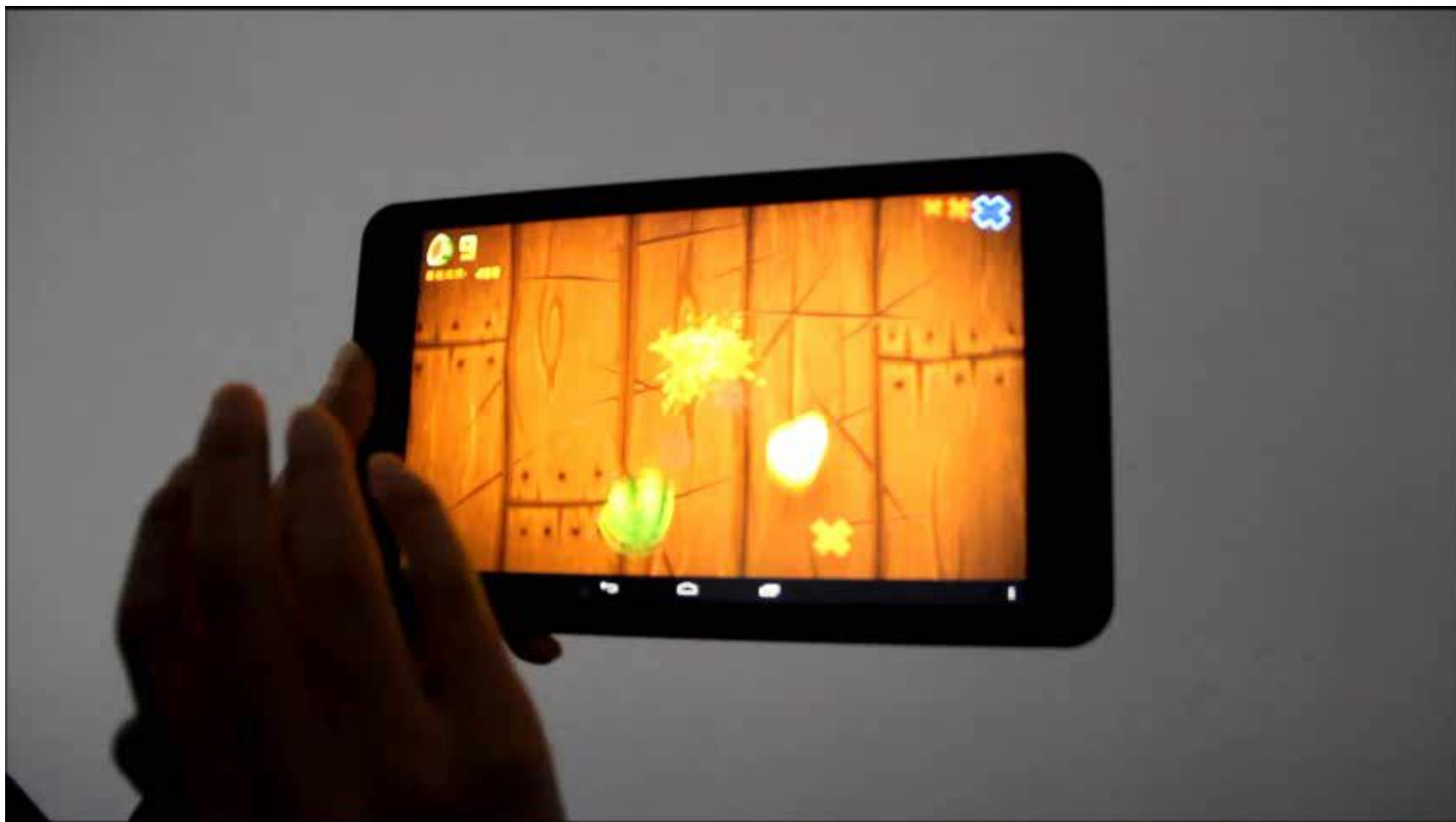
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multi-touch 7.0 inch graphene-film touch screen



Leading the Carbon Era with High Quality of Graphene

以高质量的石墨烯，创碳时代的领导者

THANKS

地址：江苏省常州市武进经济开发区西太湖大道9号-8号厂房

电话：0519-81230981 传真：0519-81230998





The Sixth Element Inc.

企业资质



已申请30余项发明专利，其中4项已授权

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政府助推企业裂变成长



2011年5月
局常委李长
理瞿研博士



2014年
司高云虎

司陈永自副司长视察我公司

斗工集团
总经理曹建国到我公司考察



苏省副省
司总经理
的汇报



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The Sixth Element Inc.

Our costumers and collaborators



中国科学院苏州纳米技术与纳米仿生研究所
Suzhou Institute of Nano-Tech and Nano-Bionics (SINANO), Chinese Academy of Sciences





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环氧防腐底漆性能比较

传统的富锌环氧底漆

- × 涂层厚 $\geq 60 \mu\text{m}$
- × 析出大量锌盐，影响层间接合力影响施工
- × 影响焊接，易出现孔洞
- × 产生大量氧化锌雾气，污染环境

有机锌-石墨烯复合底漆

- √ 涂层薄 $\leq 50 \mu\text{m}$
- √ 锌含量低，对层间接合力无影响
- √ 减少氧化锌雾气，保护环境
- √ 耐盐雾时间更长
- √ 单位面积涂布成本更低

